ON/OFF POWER SWITCH EXERCISE

1.0 Learning Outcomes.....1
2.0 Hardware Set-Up.....1
3.0 Simulink Set-Up and Switch Testing.....2
4.0 Concluding Remarks.....3
Recommended Citation.....3

1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Use the Simulink Support Package for Arduino Hardware to develop and deploy an algorithm for a simple circuit containing an LED and an on/off power switch.	☐
2	Configure and use an Arduino Uno digital (PWM) output pin to control an LED in combination with an on/off power switch.	☐

2.0 Hardware Set-Up

The hardware set-up is similar to the one used in the LED (Digital) and LED (PWM) Exercises. The exercise involves connecting an LED in series with an on/off power switch, so that the switch controls whether the LED is on or off, as shown in Figure 1.

Note: While the ACE-Box can be used for this exercise, it is not required. Only the individual components listed below are needed.

Required hardware

- ATmega328P microcontroller (Arduino Uno)
- USB Type-B cable
- Breadboard
- LED
- 220 Ω resistor
- 2 × male-to-male breadboard wires
- On/off power switch

Hardware Assembly Steps

1. Insert the on/off power switch into the breadboard. Use a male-to-male jumper wire to connect Pin 9 on the Arduino to the same column as one terminal of the switch.

2. Insert a 220 Ω resistor so that one end is in the same column as the opposite terminal of the switch. Place the other end of the resistor in a different row of the breadboard.
3. Connect the long leg (anode) of the LED to the free end of the resistor. Complete the circuit by connecting the LED's short leg (cathode) to a GND pin on the Arduino using a male-to-male jumper wire.

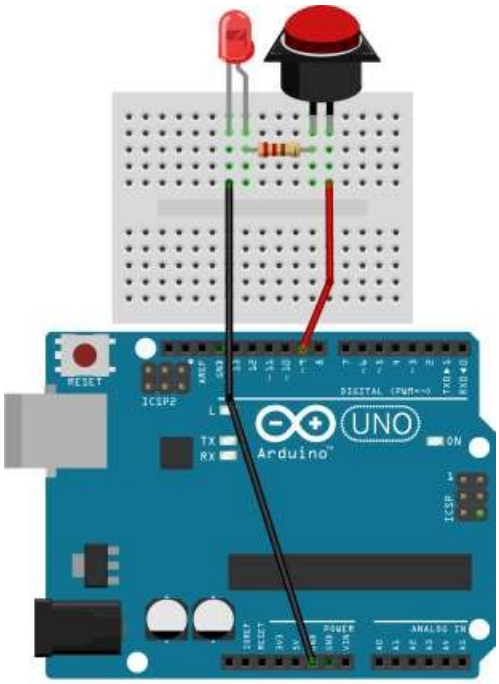


Figure 1: *Hardware Set-Up for the On/Off Power Switch Exercise*

3.0 Simulink Set-Up and Switch Testing

The on/off power switch adds no inputs or outputs to the Arduino, so the Simulink model is the same as in the LED (PWM) Exercise, as shown in Figure 2. For completeness, the full set-up is given again.

Simulink Model Construction Steps

1. From Simulink → Sources, add a Sine Wave block and connect it to an Abs (absolute value) block from Math Operations.
2. From Math Operations, connect a Slider Gain block. The lower (Low) and upper (High) limits of the gain constrain the output. For example, to limit the output to 2.3 V: 256 × 2.3 = 117.76 ≈ 117, so High would be set to 117. For this exercise, set High to 255 (equivalent to 5 V on the Arduino Uno).
3. From Simulink Support Package for Arduino Hardware → Common, add a PWM block, set the Pin number to 9, and connect three Scope blocks (from Sinks) to monitor the signals.
4. Deploy the model to the Arduino Uno and confirm that the LED brightness varies. Explore the effect of the gain value in the Slider Gain block.

To recap how to run Simulink code generation and deploy the model to the Arduino Uno, see Section 4.0 of the LED (Digital) Exercise or visit Code Generation.

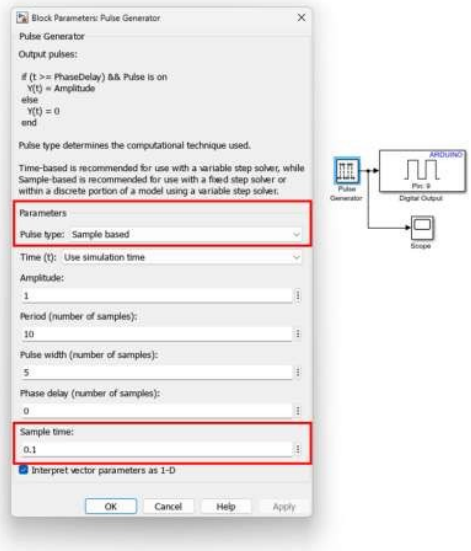


Figure 2: *Simulink Set-Up for the On/Off Power Switch Exercise*

With the model running, operate the on/off power switch and confirm that it turns the LED on and off.

4.0 Concluding Remarks

This exercise has shown how a physical on/off power switch can be combined with a PWM-controlled LED, and how a Simulink model is deployed to the ATmega328P microcontroller using the Simulink Support Package for Arduino Hardware. The switch acts directly on the circuit, while Simulink sets the brightness of the LED through PWM.

Before completing the exercise, return to the Learning Outcomes in Table 1 and confirm that you are confident you can demonstrate each one. If so, tick the corresponding Achieved box.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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